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Project Title.:

"Unlocking Brazilian E-Commerce:
Customer Behavior, Delivery Patterns & Sales Trends"

Project Objective:

To build an interactive Tableau Public dashboard that visually narrates insights uncovered through advanced exploratory analysis conducted in Python, using the Olist Brazilian E-Commerce dataset. The dashboard will be structured to support business decision-making related to customer satisfaction**,** delivery logistics, and sales optimization**.**

Key Research Questions & Hypotheses:

1. Customer & Order Behavior

- **Q1:** What are the dominant customer segments based on location, order frequency, and product categories?
    - **Location:** most customers from Southeast BR – specially SP, RJ & MG.
    - **Order frequency:** over 70% of customers made only 1 purchase – long tail behaviour
    - Popular categories: healthy_beauty, bed_bath_table and furniture_decor.

2. Sales & Category Trends

- **Q3:** Which product categories generate the most revenue and how do they perform over time?
    - **Top revenue categories:** healthy_beauty, bed_bath_table & computer_acessorie

- **Q4:** How does seasonality affect sales volumes across different product categories?
  - **Seasonal spikes:** clear November and December due to Black Friday and Holidays
  - **Sharper peaks:** some categories as Toys and Electronics

3. Delivery Performance

- **Q5:** What are the regional differences in delivery time, and how do they impact customer review scores?
  - **North & Northeast regions:** significantly longer average delivery times
  - **Longer delays related to Lower review scores**
  - **Urban centers:** SP and RJ faster deliveries positive reviews

- **Q6:** Can delivery delays be predicted based on product, seller, and order features?
  - **Top predictive features from Python**: shipping_distance, product_weight_g, seller_city.
  - Random forest classifier had average of 78% accuracy in predicting delays

4. Review & Sentiment Insights

- **Q7:** What factors contribute to negative review scores?
  - **Main Factor:** delivery delays

- **Q8:** Is there a correlation between review scores and delivery time or payment method?
  - **Yes**
  - There is no strong link found between review score and payment method

Data Source

Dataset Name: Brazilian E-Commerce Public Dataset by Olist

Source: [Kaggle - Brazilian E-Commerce Public Dataset by Olist]

Summary:

This dataset is provided by Olist, a Brazilian e-commerce company that operates a marketplace platform allowing small businesses to sell products through major online retail channels. The dataset contains detailed information about orders made on the Olist store from 2016 to 2018. It includes a comprehensive set of interrelated CSV files capturing the complete sales lifecycle, including customer data, order details, product information, payment transactions, seller profiles, and customer reviews.

Files Included:

- olist_orders_dataset.csv – order status, purchase timestamp, delivery details
- olist_customers_dataset.csv – customer location and unique identifiers
- olist_order_items_dataset.csv – product details per order
- olist_products_dataset.csv – product attributes like category, name length, etc.
- olist_sellers_dataset.csv – seller location and IDs
- olist_order_payments_dataset.csv – payment method and installments
- olist_order_reviews_dataset.csv – customer review scores and comments
- product_category_name_translation.csv – English translation of product category names

Use Case:

The dataset is ideal for a wide range of e-commerce analytics tasks such as customer segmentation, delivery performance analysis, product recommendation systems, seller performance evaluation, and sentiment analysis from customer reviews. It serves as a rich resource for both descriptive and predictive data modeling.

Size and Format:

The data is provided in CSV format and totals approximately 10 files, with over 100,000 orders and multiple data points across customers, products, and reviews.

License:

The dataset is publicly available under the terms set by the Kaggle platform and the Olist company.